

MLOPS Lab 2 Exercises

```
Dockerfile > ...
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
5 # Install EXACTLY what training used -- this is the whole point
6 COPY model_export/requirements.txt .
7 RUN pip install --no-cache-dir -r requirements.txt
8
9 COPY model_export/ /app/model_export/
10 COPY score_batch.py .
11
12 ENTRYPOINT ["python", "score_batch.py"]
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
=> exporting to image
=> => exporting layers
=> => writing image sha256:338855af6f56c1d80b1faabb3d5b8be63cf4ee61da9d6df66908220a691c0866
=> => naming to docker.io/library/cartvista-scorer:v1
• (.venv) (base) s2025ct05032@ip-10-3-22-254:~/workspace/cartvista-mlops$ docker images cartvista-scorer
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
cartvista-scorer    v1              338855af6f56   50 seconds ago  874MB
• (.venv) (base) s2025ct05032@ip-10-3-22-254:~/workspace/cartvista-mlops$
```

```
• (.venv) (base) s2025ct05032@ip-10-3-22-254:~/workspace/cartvista-mlops$ # from the cartvista-mlops folder:
docker run --rm -v "${PWD}:/data" cartvista-scorer:v1 /data/churn_data.csv /data/scored.csv
2026/10/01 13:57:16 WARNING mlflow.pyfunc: The version of Python that the model was saved in, 'Python 3.12.9', differs from the version of Python that is currently running, 'Python 3.11.16'
nd may be incompatible
Scored 5000 customers -> /data/scored.csv (predicted churners: 18)
• (.venv) (base) s2025ct05032@ip-10-3-22-254:~/workspace/cartvista-mlops$
```

DAG: cartvista_weekly_retrain

Schedule: None Next Run ID: None

10/01/2026 03:08:43 PM All Run Types All Run States Clear Filters Auto-refresh 25

Press **shift** + **f** for Shortcuts

deferred failed queued removed restarting running scheduled shutdown skipped success up_for_reschedule up_for_retry upstream_failed no_status

Run DAG: cartvista_weekly_retrain / 2026-10-01, 14:58:14 UTC / promote Clear task Mark state as... Filter DAG by task

Details Graph Gantt Code Audit Log Logs XCom Task Duration

Duration: 00:02:20

refresh_data train_logreg train_rf validate_best promote

refresh_data success @task

train_rf success @task

train_logreg success @task

validate_best success @task

promote success @task

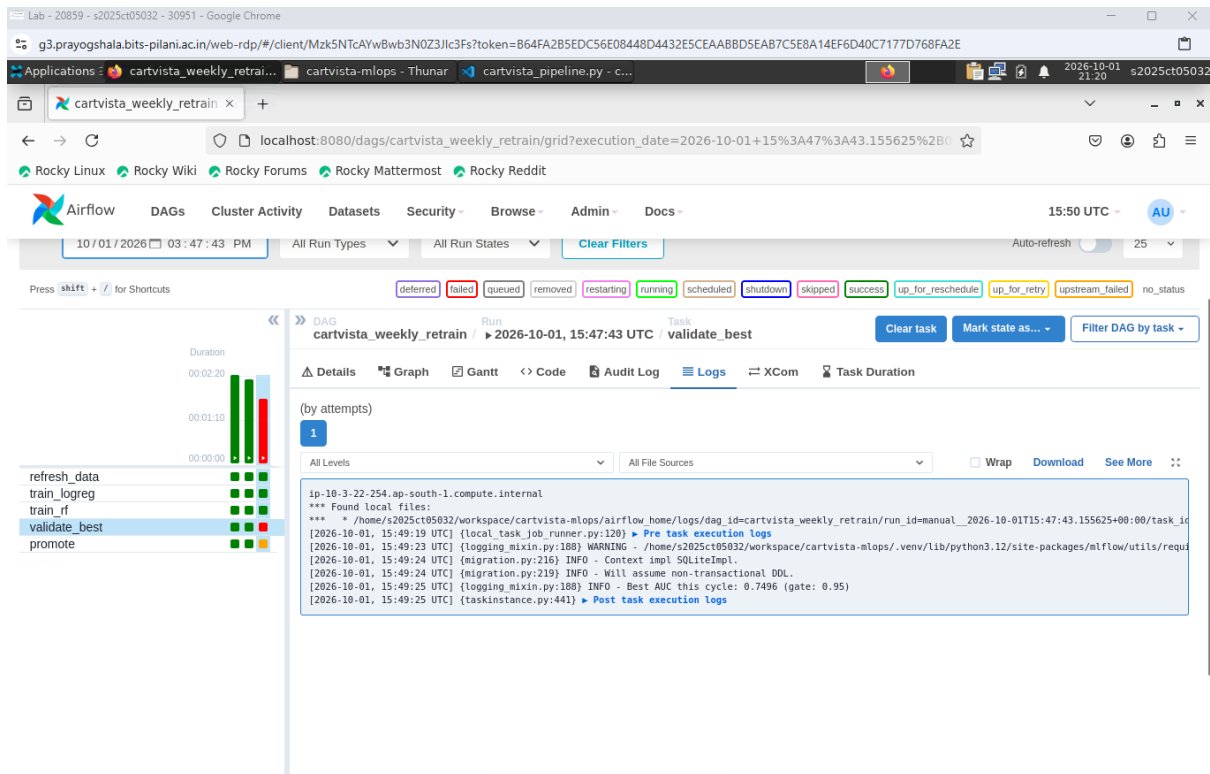
Layout: Left -> Right

List Dag Run

Search

Actions Record Count: 1

	State	Dag Id	Logical Date	Run Id	Run Type	Queued At	Start Date	End Date
☐	☐ ☐ ☐ failed	cartvista_weekly_retrain	2026-10-01, 15:47:43	manual__2026-10-01T15:47:43.155625+00:00	manual	2026-10-01, 15:47:43	2026-10-01, 15:47:45	2026-10-01, 15:49:21



Exercises:

E1 — Retry policy

Ans. Retries are relatively safe for model-training tasks because running the training task again with the same input data and configuration produces the same intended outcome and does not cause an irreversible business action. A task such as sending retention emails is not safely retryable unless it is designed to be idempotent, because a retry could send the same email multiple times to the same customer.

E2 — Shrink the image

Ans. The original **cartvista-scorer:v1** Docker image size was **874 MB**. I optimized the inference container by removing unnecessary training and MLflow runtime dependencies and retaining only the libraries required for model inference. I also removed unnecessary cache, bytecode, and other non-runtime files from the image.

After optimization, the **cartvista-scorer:v2** image size was reduced to **348 MB**, which is below the required **450 MB** limit.

Before: 874 MB

After: 348 MB

Reduction: 526 MB (~60.2%)

The optimized container was then executed using the same batch-scoring workflow as Step A4 to verify that the model continued to generate predictions correctly. Therefore, the image size was significantly reduced without breaking inference functionality.

```
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ docker images c
artvista-scorer
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
cartvista-scorer    v2           970cff06fa79     30 seconds ago   348MB
cartvista-scorer    v1           21a782fcc008     12 minutes ago   874MB
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ docker run --rm
\
-v "$(pwd):/data" \
cartvista-scorer:v2 \
/data/churn_data.csv \
/data/scored_v2.csv
Scored 5000 customers -> /data/scored v2.csv (predicted churners: 18)
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ diff scored.csv
scored_v2.csv
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$
```

E3 — Schedule it

Ans. I changed the DAG schedule from None to `schedule="0 2 * * 1"`

The cron expression `0 2 * * 1` schedules the CartVista retraining pipeline to execute every Monday at 02:00.

The next three scheduled runs displayed by Airflow were:

- 05-Oct-2026 at 02:00
- 12-Oct-2026 at 02:00
- 19-Oct-2026 at 02:00

`catchup=False` is appropriate because the pipeline should process current and future weekly retraining cycles rather than automatically executing all historical weekly runs since the DAG's start date. Enabling `catchup` could result in unnecessary retraining jobs and duplicate MLflow runs/model versions.

Your final DAG decorator should therefore be:

```
@dag( schedule="0 2 * * 1", start_date=datetime(2026, 1, 1), catchup=False,
tags=["cartvista"])
```

The screenshot displays the Airflow web interface for the DAG 'cartvista_weekly_retrain'. The top panel shows the DAG's code, which includes a Python function 'run(cmd)' and a DAG definition with a weekly schedule. The middle panel shows the DAG's details, including the next run time and a table of task dependencies. The bottom panel shows a bar chart of task durations.

```

PROJECT = pathlib.Path(__file__).resolve().parents[2] # cartvista-mlops
AUC_GATE = 0.95 # a model below this bar must NOT become champion

def run(cmd):
    r = subprocess.run(cmd, cwd=PROJECT, shell=True,
                        capture_output=True, text=True)
    print(r.stdout, r.stderr)
    r.check_returncode()
    return r.stdout

@dag(
    schedule="0 2 * * 1",
    start_date=datetime(2026, 1, 1),
    catchup=False,
    tags=["cartvista"]
)

```

The DAG details panel shows the following information:

- Next dagrun create after: 2026-10-05T02:00:00+00:00
- Next dagrun data interval end: 2026-10-05T02:00:00+00:00
- Next dagrun data interval start: 2026-09-28T02:00:00+00:00
- Pickle id: null
- Root dag id: null
- Scheduler lock: null
- Timetable description: At 02:00, only on Monday

The task dependencies table shows the following tasks and their dependencies:

Task	Dependencies
refresh_data	
train_logreg	refresh_data
train_rf	refresh_data
validate_best	train_logreg, train_rf
promote	validate_best

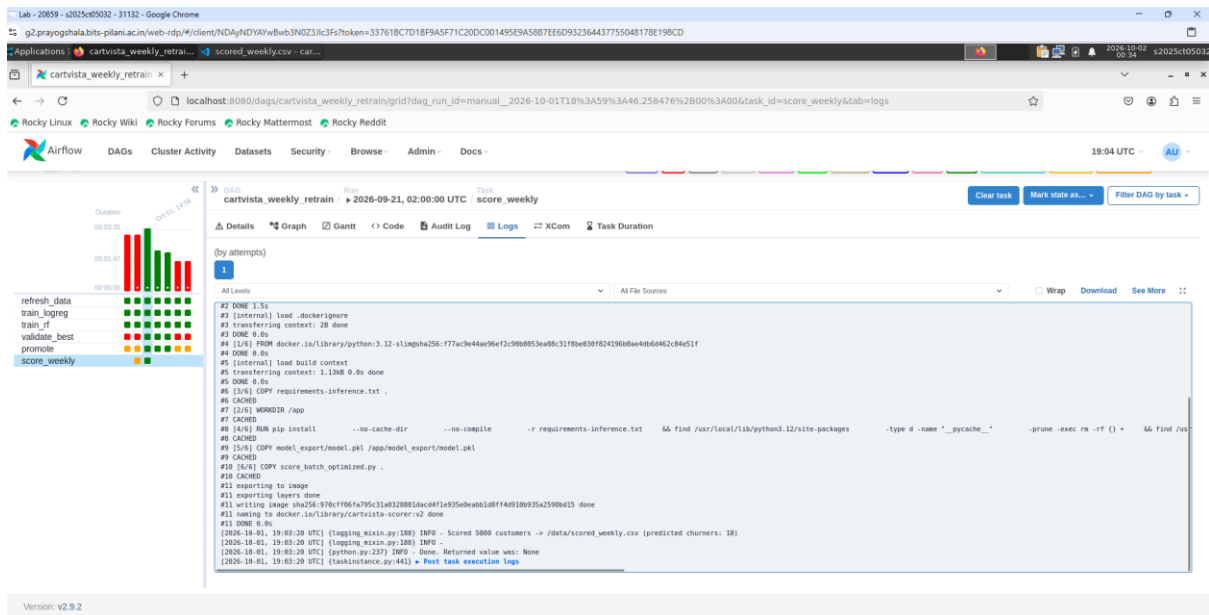
E4 — Containerized task

Ans. I added a final score_weekly task after the model promotion step. Once the candidate model passes validate_best, the promote task moves the @champion alias to the approved model. The score_weekly task then executes export_champion.py to export the newly promoted champion, rebuilds the Docker scorer image so that the new model is packaged inside it, and runs the container against churn_data.csv. The resulting scored_weekly.csv contains the churn_pred predictions.

The final pipeline is:

refresh_data → [train_logreg, train_rf] → validate_best → promote → score_weekly

This closes the ML lifecycle because the pipeline now produces a usable batch-scoring artifact rather than ending only with a new model version in the registry.



```
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ ls -lh scored_weekly.csv
-rw-r--r--. 1 root root 114K Oct  2 00:33 scored_weekly.csv
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ head scored_weekly.csv
tenure,months,monthly_spend,orders_per_month,support_tickets,uses_discounts,app_sessions,churn,churn_pred
7,294.86,3,1,1,8,0,0
55,2112.44,7,2,1,15,0,0
47,2861.39,1,0,1,15,0,0
32,2521.62,5,0,0,18,0,0
31,976.15,1,2,1,13,0,0
61,349.31,2,3,1,16,0,0
7,2832.98,4,2,1,14,0,0
50,2823.78,1,2,0,16,0,0
15,1690.76,1,3,0,14,1,0
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$
```

E5 (stretch) — Failure alert

Ans. I added an `on_failure_callback` to the `validate_best` Airflow task. When validation fails, the callback obtains the task exception, extracts the failing AUC value, and appends the timestamp, AUC, and failure information to `gate_failures.log`.

To test the callback, I temporarily increased `AUC_GATE` from 0.72 to 0.95. Since the best candidate model's AUC was below this threshold, `validate_best` failed. Airflow invoked the failure callback and created an entry in `gate_failures.log`. Because `promote` and `score_weekly` are downstream of the validation task, they did not execute after the gate failure.

This provides a persistent audit record of model-quality failures and helps identify when and why a model was prevented from being promoted.

```

def gate_failure_callback(context):
    """Write failed validation details to gate_failures.log."""

    exception = str(context.get("exception", ""))

    # Extract AUC from:
    # "Quality gate FAILED: 0.7561 < 0.95"
    match = re.search(
        r"Quality gate FAILED:\s*([0-9.]+)",
        exception
    )

    failing_auc = match.group(1) if match else "unknown"

    log_file = PROJECT / "gate_failures.log"

    timestamp = datetime.now().isoformat()

    with open(log_file, "a") as f:
        f.write(
            f"{timestamp} "
            f"AUC={failing_auc} "
            f"ERROR={exception}\n"
        )

    print(
        f"Failure recorded in {log_file}: "
        f"AUC={failing_auc}"
    )

```

```

@task(on_failure_callback=gate_failure_callback)
def validate_best() -> float:
    import mlflow
    from mlflow import MlflowClient
    mlflow.set_tracking_uri(f"sqlite:/// {PROJECT}/mlflow.db")
    client = MlflowClient()
    exp = client.get_experiment_by_name("cartvista-churn")
    best = client.search_runs([exp.experiment_id],
        order_by=["metrics.test_auc DESC"], max_results=1)[0]
    auc = best.data.metrics["test_auc"]
    print(f"Best AUC this cycle: {auc:.4f} (gate: {AUC_GATE})")
    if auc < AUC_GATE:
        raise ValueError(f"Quality gate FAILED: {auc:.4f} < {AUC_GATE}")
    return auc

```

```

(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ ls -lh gate_failures.log
-rw-r--r-- 1 s2025ct05032 s2025ct05032 79 Oct  2 00:42 gate_failures.log
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$ cat gate_failures.log
2026-10-02T00:42:36.711602 AUC=0.7496 ERROR=Quality gate FAILED: 0.7496 < 0.95
(.venv) (base) s2025ct05032@ip-10-3-17-157:~/workspace/cartvista-mlops$

```

Lab - 20859 - a2025c0502 - 31132 - Google Chrome

g2.prayogshala.bity-pilani.ac.in/web-rtsp/#/client/NDAYvWeb3N0Z3Rlc3Fs/token=337618CD7D18F9A5F71C2D0C001495E8A5887E6D932364437755048178E198CD

Applications: cartvista_weekly_retrain... cartvista_pipeline.py - C...

cartvista_weekly_retrain: x

localhost:8080/dags/cartvista_weekly_retrain?tab=logs&task_id=validate_best&dag_run_id=manual__2026-10-01T19%3A09%3A46.338148%2B00%3A00

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19:15 UTC AU

Duration: 00:00:38, 00:01:47, 00:00:00

refresh_data train_logreg train_rf validate_best promote score_weekly

DAG: cartvista_weekly_retrain Run: 2026-09-21, 02:00:00 UTC Task: validate_best

Clear task Mark state as... Filter DAG by task

Details Graph Gantt Code Audit Log Logs XCom Task Duration

(by attempts)

All Levels All File Sources Wrap Download See More

```
File: "/home/s2025c05032/workspace/cartvista-nlops/.venv/lib/python3.12/site-packages/airflow/models/taskinstance.py", line 432, in _execute_callable
return execute_callable(context=context, **execute_callable_args)
File: "/home/s2025c05032/workspace/cartvista-nlops/.venv/lib/python3.12/site-packages/airflow/models/baseoperator.py", line 491, in wrapper
return func(self, *args, **kwargs)
File: "/home/s2025c05032/workspace/cartvista-nlops/.venv/lib/python3.12/site-packages/airflow/decorators/base.py", line 265, in execute
return_value = super().execute(context)
File: "/home/s2025c05032/workspace/cartvista-nlops/.venv/lib/python3.12/site-packages/airflow/models/baseoperator.py", line 491, in wrapper
return func(self, *args, **kwargs)
File: "/home/s2025c05032/workspace/cartvista-nlops/.venv/lib/python3.12/site-packages/airflow/operators/python.py", line 235, in execute
return_value = self.execute_callable()
File: "/home/s2025c05032/workspace/cartvista-nlops/.venv/lib/python3.12/site-packages/airflow/operators/python.py", line 252, in execute_callable
return self.python_callable(*self.op_args, **self.op_kwargs)
File: "/home/s2025c05032/workspace/cartvista-nlops/airflow/home/dags/cartvista_pipeline.py", line 85, in validate_best
raise ValueError(f"Quality gate FAILED: {spec.ATL} < {AUC_GATE}")
ValueError: Quality gate FAILED: 0.7496 < 0.95
[2026-10-01, 19:12:36 UTC] [taskinstance.py:1385] INFO - Marking task as FAILED. dag_id=cartvista_weekly_retrain, task_id=validate_best, run_id=manual__2026-10-01T19:09:46.338148-00:00, execution_date=20261001T190946, start_date=20261001T191236
[2026-10-01, 19:12:36 UTC] [logging_mixin.py:188] INFO - Failure recorded in /home/s2025c05032/workspace/cartvista-nlops/gate_failures.log: AUC=0.7496
[2026-10-01, 19:12:36 UTC] [standard_task_runner.py:181] ERROR - Failed to execute job 40 for task validate_best (Quality gate FAILED: 0.7496 < 0.95): 23566
[2026-10-01, 19:12:37 UTC] [local_task_job_runner.py:240] INFO - Task exited with return code 1
[2026-10-01, 19:12:37 UTC] [taskinstance.py:3503] INFO - 0 downstream tasks scheduled from follow-on schedule check
[2026-10-01, 19:12:37 UTC] [local_task_job_runner.py:222] *** Log group end
```

Version: v2.9.2

localhost:8080/cluster_activity 14261391273d307aaf537cc78277c3d1

next ^ v Highlight All Match Case Match Diacritics Whole Words 1 of 5 matches Reached end of page, continued from top